5.7 Classwork: Analytic Geometry (no calculator) — Markscheme

1. A straight line L passes through the points $A(6,0)$ and $B(0,-4)$.

(a) Find the gradient of L. [2]

$$m = \frac{0-(-4)}{6-0} = \frac{4}{6}$$ **(M1)**

$$= \frac{2}{3}$$ **(A1)**

(b) Write down the y-intercept of L. [1]

y-intercept $= -4$ **(A1)**

(c) Find the equation of L in the form $ax + by + d = 0$, where a, b, $d \in \mathbb{Z}$. [2]

$y = \frac{2}{3}x - 4$ **(M1)**

$3y = 2x - 12$

$2x - 3y - 12 = 0$ **(A1)**

(d) A second line L_2 is perpendicular to L and passes through the point A. Find the equation of L_2. [3]

Perpendicular gradient: $m_2 = -\frac{3}{2}$ **(A1)**

$y - 0 = -\frac{3}{2}(x-6)$ **(M1)** **(A1)**

$y = -\frac{3}{2}x + 9$

2. The graph of $f(x) = x^2 - 6x + 5$ is a parabola.

(a) Find the x-intercepts of the graph of f. [2]

$x^2 - 6x + 5 = 0$ **(M1)**

$(x-1)(x-5) = 0$

$x = 1$ and $x = 5$ **(A1)**

(b) Write $f(x)$ in the form $(x-h)^2 + k$. [2]

$f(x) = (x^2 - 6x + 9) - 9 + 5$ **(M1)**

$= (x-3)^2 - 4$ **(A1)**

(c) Hence write down the coordinates of the vertex of the parabola. [1]

Vertex $= (3, -4)$ **(A1)**

(d) The graph of f is translated by the vector $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$. Find the equation of the new graph in the form $y = x^2 + bx + c$. [4]

New vertex: $(3+3,\ -4+2) = (6, -2)$ **(A1)(A1)**

$y = (x-6)^2 - 2$ **(M1)**

$= x^2 - 12x + 36 - 2$

$= x^2 - 12x + 34$ **(A1)**

3. The line $y = 2x + k$ is tangent to the parabola $y = x^2 - 4x + 7$.

(a) Write down an equation whose solutions are the x-coordinates of the points of intersection of the line and the parabola. [2]

$2x + k = x^2 - 4x + 7$ **(M1)**

$x^2 - 6x + (7 - k) = 0$ **(A1)**

(b) Hence show that $x^2 - 6x + (7 - k) = 0$. [1]

$x^2 - 4x + 7 - 2x - k = 0$ **(A1) (AG)**

$x^2 - 6x + (7 - k) = 0$

(c) State the condition on the discriminant for the line to be tangent to the parabola. [1]

$\Delta = 0$ (the quadratic has exactly one solution) **(A1)**

(d) Find the value of k. [3]

$\Delta = b^2 - 4ac = (-6)^2 - 4(1)(7 - k)$ **(M1)**

$= 36 - 28 + 4k = 8 + 4k$ **(A1)**

$8 + 4k = 0 \Rightarrow k = -2$ **(A1)**

(e) Find the coordinates of the point of tangency. [3]

$x^2 - 6x + 9 = 0$ **(M1)**

$(x - 3)^2 = 0 \Rightarrow x = 3$ **(A1)**

$y = 2(3) + (-2) = 4$

Point of tangency: $(3, 4)$ **(A1)**

4. A ball is thrown vertically upward. Its height h metres above the ground at time t seconds is modelled by

$$h(t) = -5t^2 + 20t + 1, \quad \text{for } t \geq 0.$$

The velocity of the ball at time t is given by the gradient of the tangent to the curve $h(t)$. For a quadratic $h(t) = at^2 + bt + c$, the gradient of the tangent at t is $h'(t) = 2at + b$.

(a) Write down the height of the ball when $t = 0$. [1]

$h(0) = -5(0) + 20(0) + 1 = 1$ m **(A1)**

(b) Find $h'(t)$, the velocity of the ball at time t. [2]

$a = -5$, $b = 20$ **(M1)**

$h'(t) = 2(-5)t + 20 = -10t + 20$ **(A1)**

(c) Find the time when the ball reaches its maximum height. [2]

Maximum when $h'(t) = 0$: **(M1)**

$-10t + 20 = 0 \Rightarrow t = 2$ s **(A1)**

(d) Find the maximum height. [2]

$h(2) = -5(4) + 20(2) + 1$ **(M1)**

$= -20 + 40 + 1 = 21$ m **(A1)**

(e) Find the time when the ball hits the ground. [2]

$h(t) = 0$: $\quad -5t^2 + 20t + 1 = 0$ **(M1)**

$5t^2 - 20t - 1 = 0$

$$t = \frac{20 \pm \sqrt{400 + 20}}{10} = \frac{20 \pm \sqrt{420}}{10}$$

(A1)

$$t = 2 + \frac{\sqrt{105}}{5} \text{ s} \quad (t > 0)$$